

```

# =====
# COMPUTER VISION CONCEPT MAPS - 10 QUESTIONS
# Install Graphviz first:
# pip install graphviz
# Also install Graphviz software:
# https://graphviz.org/download/
# =====

from graphviz import Digraph

# -----
# COMMON FUNCTION
# -----
def create_map(title, filename, nodes, edges):
    dot = Digraph(name=filename, format="png")

    dot.attr(
        label=title,
        labelloc="t",
        fontsize="20",
        fontname="Arial",
        rankdir="TB",
        bgcolor="white"
    )

    dot.attr(
        "node",
        shape="box",
        style="rounded,filled",
        fillcolor="lightblue",
        color="black",
        fontname="Arial",
        fontsize="12"
    )

    dot.attr(

```

```
        "edge",
        color="black",
        arrowsize="0.8",
        fontname="Arial"
    )

    # Add nodes
    for node_id, node_text in nodes.items():
        dot.node(node_id, node_text)

    # Add connections
    for source, destination, relationship in edges:
        dot.edge(source, destination, label=relationship)

    # Save image
    dot.render(filename, cleanup=True)

    print(filename + ".png created successfully!")

# =====
# QUESTION 1
# Real-World Scene Capture to Digital Image Representat
# =====

nodes1 = {
    "A": "Real-World Scene",
    "B": "Imaging System",
    "C": "Image Sensor",
    "D": "Sampling",
    "E": "Quantization",
    "F": "Digital Image",
    "G": "Lighting",
    "H": "Noise",
    "I": "Resolution"
}

edges1 = [
    ("A", "B", "captured by"),
```

```

    ("B", "C", "focuses light on"),
    ("C", "D", "produces signal for"),
    ("D", "E", "creates samples for"),
    ("E", "F", "generates"),
    ("G", "B", "affects"),
    ("H", "C", "determines quality"),
    ("I", "F", "determines quality")
]

create_map(
    "1. Real-World Scene to Digital Image Representation",
    "question_1",
    nodes1,
    edges1
)

# =====
# QUESTION 2
# Major Stages of Visual Data Handling
# =====

nodes2 = {
    "A": "Image Acquisition",
    "B": "Preprocessing",
    "C": "Feature Extraction",
    "D": "Mid-Level Analysis",
    "E": "High-Level Interpretation",
    "F": "Meaningful Output",
    "G": "Knowledge and Models"
}

edges2 = [
    ("A", "B", "provides image to"),
    ("B", "C", "improves image for"),
    ("C", "D", "provides features to"),
    ("D", "E", "supports"),
    ("E", "F", "produces"),
    ("G", "D", "guides"),

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```
    ("G", "E", "supports")
]

create_map(
    "2. Visual Data Handling in Computer Vision",
    "question_2" Add text cell
    nodes2,
    edges2
)

# =====
# QUESTION 3
# Image Resolution and Intensity Levels
# =====

nodes3 = {
    "A": "Image Resolution",
    "B": "Spatial Resolution",
    "C": "Intensity Resolution",
    "D": "Pixel Density",
    "E": "Image Detail",
    "F": "Intensity Levels",
    "G": "Contrast and Smoothness",
    "H": "Image Quality",
    "I": "Application Usability"
}

edges3 = [
    ("A", "B", "includes"),
    ("A", "C", "includes"),
    ("B", "D", "determines"),
    ("D", "E", "improves"),
    ("C", "F", "determines"),
    ("F", "G", "improves"),
    ("E", "H", "increases"),
    ("G", "H", "increases"),
    ("H", "I", "affects")
]
```

```
create_map(  
    "3. Image Resolution and Intensity Levels",  
    "question_3",  
    nodes3,  
    edges3  
)
```

Add text cell

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# =====  
# QUESTION 4  
# Factors in Image Acquisition Conditions  
# =====
```

```
nodes4 = {  
    "A": "Image Acquisition Conditions",  
    "B": "Illumination",  
    "C": "Camera Settings",  
    "D": "Environment",  
    "E": "Brightness and Contrast",  
    "F": "Image Quality",  
    "G": "Preprocessing",  
    "H": "Feature Extraction",  
    "I": "Application Performance"  
}
```

```
edges4 = [  
    ("A", "B", "depends on"),  
    ("A", "C", "depends on"),  
    ("A", "D", "depends on"),  
    ("B", "E", "affects"),  
    ("C", "E", "affects"),  
    ("D", "E", "affects"),  
    ("E", "F", "determines"),  
    ("F", "G", "influences"),  
    ("G", "H", "supports"),  
    ("H", "I", "improves")  
]
```

```
create_map(  
    "4. Image Acquisition Conditions and Performance",  
    "question_4",  
    nodes4,  
    edges4  
)
```

Add text cell

```
# =====  
# QUESTION 5  
# System Capabilities and Vision Applications  
# =====
```

```
nodes5 = {  
    "A": "System Capabilities",  
    "B": "Sensor Quality",  
    "C": "Processing Power",  
    "D": "Algorithm Efficiency",  
    "E": "Storage and Bandwidth",  
    "F": "Application Suitability",  
    "G": "Processing Speed",  
    "H": "Accuracy and Reliability",  
    "I": "Deployment Feasibility",  
    "J": "Vision-Based Application"  
}
```

```
edges5 = [  
    ("A", "B", "includes"),  
    ("A", "C", "includes"),  
    ("A", "D", "includes"),  
    ("A", "E", "includes"),  
    ("B", "F", "determines"),  
    ("C", "G", "improves"),  
    ("D", "H", "improves"),  
    ("E", "I", "supports"),  
    ("F", "J", "selects"),  
    ("G", "J", "supports"),  
    ("H", "J", "improves"),  
    ("I", "J", "enables")
```

```
]

create_map(
    "5. System Capabilities and Vision Applications",
    "question_5",
    nodes5,
    edges5
)

# =====
# QUESTION 6
# Complete Workflow of Digital Image Formation
# =====

nodes6 = {
    "A": "Real-World Scene",
    "B": "Light Reflection",
    "C": "Image Sensing",
    "D": "Continuous Image Signal",
    "E": "Sampling",
    "F": "Discrete Pixels",
    "G": "Quantization",
    "H": "Discrete Intensity Values",
    "I": "Digital Image",
    "J": "Spatial Resolution",
    "K": "Intensity Resolution",
    "L": "Image Quality"
}

edges6 = [
    ("A", "B", "produces"),
    ("B", "C", "reaches"),
    ("C", "D", "generates"),
    ("D", "E", "converted by"),
    ("E", "F", "creates"),
    ("F", "G", "processed by"),
    ("G", "H", "creates"),
    ("H", "I", "forms"),
```

```

    ("E", "J", "determines"),
    ("G", "K", "determines"),
    ("J", "L", "affects"),
    ("K", "L", "affects")
]

Add text cell

create_map(
    "6. Workflow of Digital Image Formation",
    "question_6",
    nodes6,
    edges6
)

# =====
# QUESTION 7
# Levels of Computer Vision
# =====

nodes7 = {
    "A": "Low-Level Computer Vision",
    "B": "Mid-Level Computer Vision",
    "C": "High-Level Computer Vision",
    "D": "Filtering, Edges and Corners",
    "E": "Segmentation and Object Detection",
    "F": "Recognition, Reasoning and Decisions"
}

edges7 = [
    ("A", "B", "provides data to"),
    ("B", "C", "supports"),
    ("A", "D", "performs"),
    ("B", "E", "performs"),
    ("C", "F", "performs")
]

create_map(
    "7. Levels of Computer Vision",
    "question 7",

```

```
nodes7,
edges7
)

# ===== Add text cell =====
# QUESTION 8
# Factors Affecting Spatial Resolution
# =====

nodes8 = {
    "A": "Spatial Resolution",
    "B": "Sampling Rate",
    "C": "Pixel Density",
    "D": "Sensor Size",
    "E": "Distance to Scene",
    "F": "Image Detail",
    "G": "Application Requirements",
    "H": "Medical Imaging",
    "I": "Remote Sensing",
    "J": "Surveillance"
}

edges8 = [
    ("B", "A", "affects"),
    ("C", "A", "affects"),
    ("D", "A", "affects"),
    ("E", "A", "affects"),
    ("A", "F", "determines"),
    ("F", "G", "supports"),
    ("G", "H", "used in"),
    ("G", "I", "used in"),
    ("G", "J", "used in")
]

create_map(
    "8. Factors Affecting Spatial Resolution",
    "question_8",
    nodes8,
```

```

edges8
)

# =====
# QUESTION 9
# Image Sensing Mechanisms
# =====

nodes9 = {
    "A": "Image Sensing Mechanism",
    "B": "Sensor Characteristics",
    "C": "Noise Generation",
    "D": "Pixel Size",
    "E": "Quantum Efficiency",
    "F": "Dynamic Range",
    "G": "Frame Rate",
    "H": "Shot Noise",
    "I": "Thermal Noise",
    "J": "Read Noise",
    "K": "Acquisition Quality",
    "L": "Final Image Quality"
}

edges9 = [
    ("A", "B", "depends on"),
    ("A", "C", "may generate"),
    ("B", "D", "includes"),
    ("B", "E", "includes"),
    ("B", "F", "includes"),
    ("B", "G", "includes"),
    ("C", "H", "includes"),
    ("C", "I", "includes"),
    ("C", "J", "includes"),
    ("D", "K", "affects"),
    ("E", "K", "improves"),
    ("F", "K", "improves"),
    ("G", "K", "affects"),
    ("H", "K", "reduces"),

```

```
    ("I", "K", "reduces"),
    ("J", "K", "reduces"),
    ("K", "L", "determines")
]

create_map(
    "9. Role of Image Sensing Mechanisms",
    "question_9",
    nodes9,
    edges9
)
```

```
# =====
# QUESTION 10
# Sampling, Aliasing and Image Distortion
# =====
```

```
nodes10 = {
    "A": "Continuous Image",
    "B": "Sampling",
    "C": "Sampling Rate Adequate?",
    "D": "Accurate Image Representation",
    "E": "Aliasing",
    "F": "False Patterns",
    "G": "Jagged Edges",
    "H": "Image Distortion",
    "I": "Reduced Image Quality"
}
```

```
edges10 = [
    ("A", "B", "converted using"),
    ("B", "C", "checks"),
    ("C", "D", "Yes: Nyquist rate"),
    ("C", "E", "No: below Nyquist"),
    ("E", "F", "causes"),
    ("E", "G", "causes"),
    ("E", "H", "causes"),
    ("F", "I", "reduces"),
```

```
    ("G", "I", "reduces"),  
    ("H", "I", "reduces")  
]  
  
create_map(  
    "10. Sampling, Add text cell, and Image Distortion",  
    "question_10",  
    nodes10,  
    edges10  
)  
  
print("\nAll 10 concept-map images are generated!")
```

```
question_1.png created successfully!  
question_2.png created successfully!  
question_3.png created successfully!  
question_4.png created successfully!  
question_5.png created successfully!  
question_6.png created successfully!  
question_7.png created successfully!  
question_8.png created successfully!  
question_9.png created successfully!  
question_10.png created successfully!
```

All 10 concept-map images are generated!

Add text cell